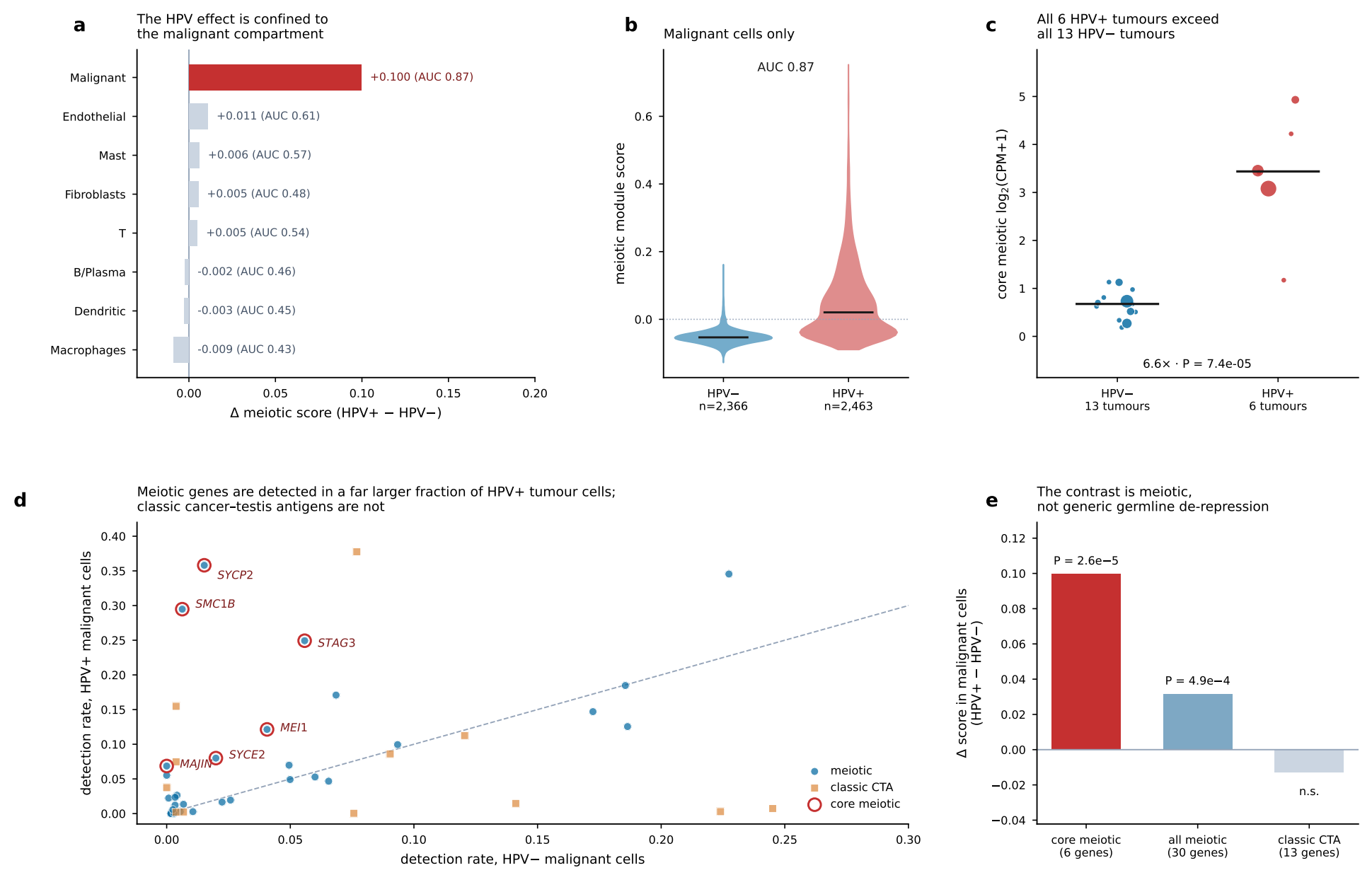
a
The HPV effect is confined to the malignant compartment
Malignant
Endothelial
Mast
Fibroblasts
T
B/Plasma
Dendritic
Macrophages
+0.100 (AUC 0.87)
+0.011 (AUC 0.61)
+0.006 (AUC 0.57)
+0.005 (AUC 0.48)
+0.005 (AUC 0.54)
-0.002 (AUC 0.46)
-0.003 (AUC 0.45)
-0.009 (AUC 0.43)
0.00
0.05
0.10
0.15
0.20
Δ meiotic score (HPV+ − HPV−)
b
Malignant cells only
AUC 0.87
meiotic module score
0.0
0.2
0.4
0.6
HPV−
n=2,366
HPV+
n=2,463
c
All 6 HPV+ tumours exceed all 13 HPV− tumours
core meiotic log2(CPM+1)
0
1
2
3
4
5
6.6× · P = 7.4e-05
HPV−
13 tumours
HPV+
6 tumours
d
Meiotic genes are detected in a far larger fraction of HPV+ tumour cells; classic cancer–testis antigens are not
detection rate, HPV+ malignant cells
detection rate, HPV− malignant cells
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0.35
0.40
SYCP2
SMC1B
STAG3
MEI1
SYCE2
MAJIN
meiotic
classic CTA
core meiotic
e
The contrast is meiotic, not generic germline de-repression
Δ score in malignant cells (HPV+ − HPV−)
P = 2.6e−5
P = 4.9e−4
n.s.
0.12
0.10
0.08
0.06
0.04
0.02
0.00
−0.02
−0.04
core meiotic
(6 genes)
all meiotic
(30 genes)
classic CTA
(13 genes)